

VALLABH SATHE

Full Stack Developer | AI & Agentic Systems

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PROFILE SUMMARY

AI-driven Full Stack Developer experienced in designing and deploying end-to-end intelligent applications. Built scalable systems using LLMs, real-time data pipelines, and cloud infrastructure. Hands-on experience from internship, projects, and hackathons with a focus on solving real-world problems.

WORK EXPERIENCE

Full Stack Developer Intern

Jun 2025 – Present

AlgoAnalytics Technologies Pvt. Ltd.

- Developed a fully responsive product showcase platform using React.js and Tailwind CSS, improving user engagement by 40% through optimized UI rendering and efficient state management
- Architected and implemented secure authentication and user management systems using AWS Lambda, API Gateway, and DynamoDB, ensuring scalable and reliable serverless backend performance
- Engineered a real-time fire and smoke detection system using a computer vision pipeline integrated with Kafka streaming, achieving low-latency data processing and high-throughput inference
- Designed and optimized YOLO-based object detection models to solve occluded road sign recognition, leveraging data augmentation and parameter tuning to improve detection accuracy under partial visibility conditions

PROJECTS

Student Course Recommendation System [\[GitHub\]](#)

Oct 2025 – Jan 2026

- Designed and implemented an AI-driven recommendation system using Retrieval-Augmented Generation (RAG) to deliver personalized learning resource recommendations.
- Developed Python-based APIs for real-time recommendations, enabling efficient interaction between AI models and frontend systems.
- Containerized services using Docker and deployed AI pipelines on AWS, ensuring scalable model serving and 99%+ system reliability.
- Implemented ML ranking algorithms combining collaborative filtering and semantic similarity scoring to improve recommendation accuracy.

Content Moderation & Community Safety Platform [\[GitHub\]](#)

Jan 2026 – Mar 2026

- Built an enterprise-grade AI content moderation platform using a multi-agent architecture (LangGraph, LangChain, Google Gemini) to automate analysis of user-generated content with explainable decision workflows — demonstrating ability to design modular, interoperable system architectures.
- Designed a ReAct (Reason-Act-Observe) decision loop with six specialized AI agents handling toxicity, policy, reputation, analysis, decision, and enforcement — analogous to intelligent, self-healing automated operations.
- Engineered a Human-in-the-Loop (HITL) review system that auto-escalates low-confidence or high-severity cases into a priority queue, enabling human override and continuous AI learning — reflecting self-service and override capability in application frameworks.

TECHNICAL SKILLS

Programming Languages: Python, Java (Data Structures & Algorithms), C++

Cloud & DevOps: Docker, Git, GitHub, AWS

Machine Learning & AI: LLMs, RAG, LangChain, LangGraph, NLP, YOLO, scikit-learn, MCP

Web Development: React.js, Node.js, Express.js, Next.js, Tailwind CSS, MongoDB, MySQL

Libraries & Tools: Hugging Face, NumPy, Pandas

EDUCATION

Bachelor of Technology in Computer Science and Engineering

Sanjay Ghodawat University, Kolhapur

May 2026

CGPA: 8.45 / 10.00

CERTIFICATIONS

- Data Analytics with Python
- Deep Learning – IIT Ropar
- Machine Learning
- React and JavaScript (Meta)